TTV: Tokenizing Test Vectorize with γ-Scaled Attention

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Abstract

We present TTV, a pipeline that turns text into tokens, tokens into vectors, and mixes them with an attention mechanism that has a learnable scale γ.

1 Tokenizing

A sentence is split into tokens $t_1, \ldots, t_n$, where each $t_i \in \{0, 1, \ldots, |\mathcal{V}|-1\}$ and $|\mathcal{V}|$ is the vocabulary size.

2 Vectorize

An embedding matrix $E \in \mathbb{R}^{|\mathcal{V}| \times d}$ maps each token to a vector:

$$v_i = E[t_i] = [v_1, v_2, \ldots, v_d]. \tag{1}$$

Adding a position vector p_i gives

$$x_i = E[t_i] + p_i, \qquad X = [x_1; x_2; \ldots; x_n] \in \mathbb{R}^{n \times d}. \tag{2}$$

3 TTV Attention

Queries, keys and values are $Q = XW_Q$, $K = XW_K$, $V = XW_V$. We define

$$\text{Attention}_{TTV}(Q, K, V) = \text{softmax}\left(\gamma \, \frac{QK^\top}{d_k}\right) V, \tag{3}$$

$$\text{softmax}(z)_j = \frac{e^{z_j}}{\sum_m e^{z_m}}. \tag{4}$$

Here γ is learned separately for each head h and starts at $\gamma_h = \sqrt{d_k}$, so TTV starts out equal to standard scaled dot-product attention, $\text{softmax}(QK^\top/\sqrt{d_k})V$. A larger γ makes the attention sharper, and a smaller γ makes it smoother.

4 Experiment

A small 2-layer, 4-head TTV model ($d = 64$) was trained on a toy sentence. It learned γ values close to their starting value of 4.0. Next step: compare TTV with standard attention on a real dataset.

5 Conclusion